

# StitchPerfect

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# Chapter 1

## Introduction

### 1.1 Background and Context

Tailoring businesses handle a large amount of customer information, including customer details, body measurements, clothing orders, and measurement history. When these records are maintained manually, it can become difficult to retrieve old measurements, update customer information, track orders, and maintain accurate records.

The proposed StitchPerfect provides a centralized software solution for managing these activities. The system allows the administrator to store and manage customer information, body measurements, and order records through a graphical user interface.

An additional feature of the system is an automatic measurement prediction module. Given basic inputs such as height and weight, the system uses a measurement dataset containing approximately 10,000 records to estimate multiple body measurements. This can assist the tailor in obtaining an initial set of measurements before manually verifying them.

### 1.2 Problem Statement

Traditional tailoring workflows often depend on handwritten registers or separate records. Such approaches can result in duplicated information, difficulty in searching previous records, errors while copying measurements, and inefficient order management.

The project addresses these problems by developing an integrated application that stores customer and tailoring records in a database and provides functionality for adding, viewing, updating, and managing these records. Furthermore, the system introduces an automated prediction feature that

uses height and weight as inputs to estimate body measurements.

## **1.3 Project Objectives**

The main objectives of the project are:

1. To develop a centralized system for managing customer information.
2. To store and manage body measurements digitally.
3. To maintain tailoring order records in a structured database.
4. To provide CRUD functionality for the major system entities.
5. To develop an automatic body-measurement prediction feature using height and weight.
6. To use a measurement dataset of approximately 10,000 rows for the prediction component.
7. To provide a simple graphical interface that can be used efficiently by a tailor or administrator.

## **1.4 Scope of the Project**

The system focuses on the management of customers, measurements, and orders within a tailoring environment. The administrator is the primary user who enters and manages records.

The prediction component is intended to provide estimated measurements from basic physical inputs. The predicted values can be reviewed and verified by the tailor before being used for actual garment preparation.

# Chapter 2

## Methodology and Design

### 2.1 System Architecture

The proposed system follows a layered application structure consisting of a graphical user interface, application logic, prediction functionality, and a database layer.

The administrator interacts with the web interface. The application processes the entered information and performs the required database operations. Customer details, measurements, and order information are stored in MySQL. The prediction module receives height and weight and produces estimated measurement values using the prepared measurement dataset.

#### 2.1.1 High-Level Design

The major components of the system are:

- **Admin Interface:** Provides access to the management functionality.
- **Customer Management:** Handles customer records and their associated information.
- **Measurement Management:** Stores and retrieves body measurements.
- **Order Management:** Maintains tailoring order information.
- **Measurement Predictor:** Estimates body measurements from height and weight.
- **Database Layer:** Stores persistent customer, measurement, and order records using MySQL.

### 2.1.2 Technical Stack

- **Frontend:** HTML, CSS, JavaScript and Bootstrap
- **Backend:** Node.js and Express.js
- **Database:** MySQL
- **Database Connectivity:** MySQL driver for Node.js
- **Prediction Component:** Body-measurement prediction model based on a prepared measurement dataset
- **Development Tools:** Visual Studio Code and Git

## 2.2 Database Design

The database provides persistent storage for the system. The major logical entities are Customer, Measurement, and Order.

### 2.2.1 Customer Entity

The Customer entity stores information required to identify and manage a customer. A customer can have measurement records and multiple order records over time.

### 2.2.2 Measurement Entity

The Measurement entity stores body measurements required for tailoring. The system can maintain measurements such as:

- Chest
- Waist
- Hips
- Shoulder
- Sleeve
- Inseam
- Neck

Additional measurement attributes can be included according to the final database design of the project.

### **2.2.3 Order Entity**

The Order entity stores information related to garments ordered by a customer. It can be used to track the order and associate it with the corresponding customer and measurements.

## **2.3 Implementation Details**

### **2.3.1 Customer Management Module**

The customer management module allows the administrator to enter customer details and maintain existing records. CRUD operations are used to create, retrieve, update, and delete records as required.

### **2.3.2 Measurement Management Module**

The measurement management module provides facilities for recording and retrieving customer measurements. Measurements can be associated with the corresponding customer so that previous records can be accessed when required.

### **2.3.3 Order Management Module**

The order management module maintains tailoring orders and their associated customer information. This helps the administrator keep track of the work that has been requested for each customer.

### **2.3.4 Automatic Measurement Prediction**

The prediction module is designed to estimate body measurements using basic inputs, particularly height and weight. The system uses a prepared dataset containing approximately 10,000 measurement records.

The prediction workflow is:

1. The administrator enters the customer's height and weight.
2. The prediction component processes these input values.
3. The trained or prepared prediction logic estimates the required body measurements.
4. The predicted measurements are displayed to the administrator.

5. The administrator can verify the values before using them for tailoring.

The prediction feature is intended as an assistance mechanism rather than a replacement for physical measurement and professional verification.

## 2.4 System Workflow

The general workflow of the application is:

1. Administrator opens the application.
2. Customer information is entered or an existing customer is selected.
3. Measurements can be entered manually or estimated using the prediction feature.
4. Measurement records are stored in the database.
5. An order is created and associated with the customer.
6. The stored information can later be retrieved and updated.

# Chapter 3

## Results and Evaluation

### 3.1 Testing Methodology

The system can be evaluated through functional and integration testing of its major modules.

- **Customer Testing:** Verify creation, retrieval, modification, and deletion of customer records.
- **Measurement Testing:** Verify that measurement records are stored and retrieved correctly.
- **Order Testing:** Verify that orders are correctly associated with customers.
- **Database Testing:** Verify successful insertion, retrieval, updating, and deletion of database records.
- **Prediction Testing:** Verify that height and weight inputs produce measurement estimates.
- **Integration Testing:** Verify communication between the web interface, application logic, prediction component, and MySQL database.

### 3.2 Functional Results

The completed system is designed to provide a single interface through which the administrator can manage the major tailoring activities.

| <b>Module</b>          | <b>Expected Result</b>                                                 |
|------------------------|------------------------------------------------------------------------|
| Customer Management    | Customer records can be added, viewed, updated, and deleted.           |
| Measurement Management | Body measurements can be stored and retrieved for customers.           |
| Order Management       | Tailoring orders can be created and associated with customers.         |
| Prediction Module      | Height and weight can be used to generate estimated body measurements. |
| Database               | Records remain available after being stored in MySQL.                  |

Table 3.1: Functional Results Summary

### 3.3 Prediction Dataset

The prediction component uses a body-measurement dataset containing approximately 10,000 rows. The dataset provides reference examples for estimating body measurements from input characteristics.

The final report should include the exact prediction-model metrics, dataset statistics, and validation results obtained during implementation. These values should be inserted after the final model has been evaluated.

### 3.4 Analysis

The system combines traditional record management with an automated prediction feature. Centralized database storage improves the organization and retrieval of customer information, while the prediction component can reduce the effort required to obtain an initial measurement estimate.

The predicted measurements should always be reviewed by the tailor because body measurements can vary between individuals even when their height and weight are similar.

# Chapter 4

## Conclusion

### 4.1 Summary of Work

The StitchPerfect provides a digital solution for managing customer information, measurements, and tailoring orders. The application uses a web interface and MySQL database to provide structured storage and management of records.

The project also incorporates an automated measurement prediction component that uses height and weight as inputs and a large body-measurement dataset as the reference for generating estimated measurements.

### 4.2 Key Findings

- A centralized database can simplify the management of tailoring records.
- Digital measurement records make previously stored customer information easier to retrieve and update.
- Automated prediction can provide an initial estimate of body measurements from basic inputs.
- Prediction results require verification before being used for actual garment production.

### 4.3 Future Work and Recommendations

Possible future improvements include:

1. Improving the prediction model by using a larger and more diverse measurement dataset.
2. Adding additional input features to improve measurement estimates.
3. Adding authentication and role-based access control.
4. Adding customer order status and notification functionality.
5. Providing invoice and bill generation.
6. Adding measurement history and comparison features.
7. Developing a web or mobile version of the system.
8. Evaluating multiple machine-learning algorithms and comparing their performance using suitable metrics.

## **4.4 Final Remarks**

The project demonstrates how software engineering, database management, graphical user interfaces, and predictive techniques can be combined to digitize a traditional tailoring workflow. The system provides a foundation for further automation and can be extended with additional management and prediction capabilities.